

Ans. no -1

$$P(E | -b, +m) \propto_E P(E, -b, +m)$$

$$= \sum_{a, j} P(E, -b, a, j, +m)$$

$$= \sum_{a, j} P(E) \times P(-b) \times P(a | E, -b) \times P(j | a) \times P(+m | a)$$

$$= P(E) \times P(-b) \times \sum_a P(a | E, -b) \times P(+m | a) \sum_j P(j | a)$$

$$= P(E) \times P(-b) \times \sum_a P(a | E, -b) \times P(+m | a) \times 1$$

Now,

$$P(a | \cancel{E, -b}, E) \times P(+m | a)$$

B	E	A	P(A B,E)
-b	+e	+a	.29
-b	+e	-a	.71
-b	-e	+a	.001
-b	-e	-a	.999

A	M	P(M A)
+a	+m	.7
-a	+m	.01

B	E	A	P(a, -b, E)
-b	+e	+a	.203
-b	+e	-a	.071
-b	-e	+a	.0007
-b	-e	-a	.00999

then,

$$\therefore P(a, -b, E) \times P(-b) = \text{[unclear]}$$

-b	+e	+a	.2028
-b	+e	-a	.0709
-b	-e	+a	.0007
-b	-e	-a	.00999

✓
After eliminating $-b$,

$+e$	$+a$	$.2028$
$+e$	$-a$	$.0709$
$-e$	$+a$	$.0007$
$-e$	$-a$	$.0093$

After summing out a ,

$+e$	$.2737$
$-e$	$.0106$

Now, $P(E) \propto$

$+e$	$.2737$
$-e$	$.0106$

After normalizing,

$+e$	$.9627$
$-e$	$.0373$

$+e$	$.0005474$
$-e$	$.0105788$

After normalizing,

$P(E|-b, +a)$

$+e$	$.049$
$-e$	$.951$

Ans.

Ans. no - 2

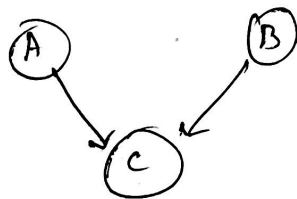
Two techniques:

- 1) Inference by enumeration
- ii) " " variable elimination

In enumeration, we first calculate (multiply) all the elements and the sum out one by one to get the result. whereas, in VE, we multiply two and then eliminate one variable; then again multiply two/more and eliminate one and so on.

Ans. no - 3

common effect:



if the common effect (C) is observed in a Bayesian network, A and B are dependent. The intuition behind this is, if $P(A|C)$ increases, $P(B|C)$ must decrease. Because they both have combined effect on C.